

## MAJOR CONTRIBUTORS TO THIS ARTICLE



**Avinash Babu**,  
senior architect, Mistral  
Solutions



**Jeff Li**,  
Overseas Department  
Asia Pacific centre  
manager, EVE Energy  
Co. Ltd



**P. Chow Reddy**,  
managing director,  
Interleaved Technologies



**Prashant  
Prabhakar Satpute**,  
sales manager,  
Powr-Gard-India,  
LiteFuse



**Rashmi Shringi**,  
business manager,  
UL India Pvt Ltd



**Sajeew Antony**,  
principal architect  
– Hardware, Dexcel  
Electronics Designs



**T. Anand**,  
principal consultant,  
Knewron

design power of just 3W. With such an efficient computing platform, one popular application is to use solar-powered wireless devices for bolstering surveillance through hidden sensors in the field.

## Ensuring reliable power source designs

A battery requires protection from different electrical conditions such as discharge, overcharge and short circuit. In the absence of adequately protected batteries, your design might turn into a ticking bomb.

The Li-ion battery pack that is used in almost all portable electronic devices, starting from a small mobile phone to a laptop, has two to three main levels of protection.

The first is the IC level of protection that is done by the battery management unit (BMU). This BMU is responsible for managing charging and discharging of the battery, and it is also the primary protection level. So if something happens and the battery goes too high on voltage or current, it would turn the pack off.

Then there is a secondary level of protection that is made compulsory by the Underwriters Laboratories (UL) test standards so as to ensure that the faults left in the primary level can be removed at the secondary level of protection. Many companies work to provide battery protection, especially for the Li-ion batteries, either at the primary level or at the secondary level.

At these different levels, the types of protections available for the Li-ion batteries are as follows:

1. PTC. A secondary level of protection that is used for automatic reset. It is considered to be a special type of fuse that protects against overcurrent and overheating.

2. CID/pressure valve. Again a secondary level of protection that disables the cells permanently in case of high pressure that may be created due to overcharging.

3. PCB protection. A primary level protection that protects against overcharging, overdischarging and even overcurrent.

Like the other types of batteries and cells, solar cells need to be protected too. Solar cells need fuses called solar fuses, which need to be certified either by UL or International Electro Commission (IEC). IEC certification is now more prominent in India. In IEC, if a fuse is tested for 135 per cent on overload, it should continue working till 60 minutes. On the other hand, in UL it should go on at 145 per cent for 60 minutes. Although certification ensures that the solar fuses are safe to be used with the solar cells, there is a requirement of a perfect material for the fuse that would make it more robust to support the cell in extreme conditions.

Industries are now coming up with a melamine-based fuse that can resist even high hot temperatures and low cold temperatures, thus allowing the solar cells to work with the same efficiency all around.

## New tools to spruce up designing

A number of tools have been introduced recently that can help simplify

the design process in numerous ways.

**Interactive battery power design tool.** This is a powerful tool that helps device developers and designers predetermine the battery power requirements for the product being designed.

This tool helps determine the parameters such as size of the battery required, battery life and energy harvesting, all with respect to the application where the device would be used.

It helps product designers and developers predetermine the battery power requirements for new products. Users are allowed to select a pre-configured product scenario or build their own use case using the defaults provided.

**Watson by Logic PD.** Watson is an application to measure power, to monitor performance and to deliver real-time feedback of the battery-powered device. It helps in minimising power and maximising the battery life for the end product by enabling the engineers to analyse data through a graphical feedback system, without having to use external oscilloscopes or digital multimeters.

**Spy on batteries.** This is a device that can help to keep a check on the actions of the lithium ions inside a nano battery. This data can, in turn, be used to develop improvised batteries that would last longer, and can also be used to power all applications – starting from an electric vehicle to a small cell phone. As lithium batteries have been so popular, this device can bring a revolutionary enhancement in the market. Designers would not have to restrict their product design because of the battery to be used. ●

*Sneha Ambastha is a technical journalist and Dilin Anand is a senior assistant editor at EBY*